

Question	Answer	Marks
5(a)	wavelength: - wavelength = distance between successive / adjacent in phase points / wavefronts / crests / troughs $\lambda = d / (\text{number of}) \text{ oscillations}$ frequency: - frequency = (number of) oscillations / cycles crests / troughs / wavefronts (passing a point) per unit time $f = (\text{number of}) \text{ oscillations} / t$ One correct point from either list	B1
	One correct point from both lists and speed = distance / time and one of: - wavelength $\times$ frequency (= distance per unit time) = speed $[(\text{number of}) \text{ oscillations} / t] \times [d / (\text{number of}) \text{ oscillations}] = f\lambda$ $v (= d / t) = \lambda / (1/f) = f\lambda$ or $v (= d / t) = \lambda / T = f\lambda$	B1
5(b)(i)	$T = 4 \times 10^{-3}$ $f = 1 / T = 1 / 0.004$	C1
	$f = 250 \text{ Hz}$	A1

Question	Answer	Marks
5(b)(ii)	$f_o = f_s v / (v - v_s)$ $250 = 236 \times v / (v - 20)$ $v = (250 \times 20) / (250 - 236)$	C1
	$= 360 \text{ m s}^{-1}$	A1

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